Manual

KPI Monitoring / OEE (MOC)
MDE-KMO 8.1

Version 1.0.8360
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Contents

1 Übersicht Kennzahlenmonitor / OEE

Purpose

The key performance indicators / OEE monitor function package makes it possible to perform a sound evaluation of the data recorded in MDE (quantities, durations) over time periods based on defined key performance indicators.

Implementation considerations

You use the function package if:

- You would like to monitor and assess the output of separate machines, machine groups or departments based on key performance indicators
- You would like to monitor and assess the output of individual machines, machine groups or departments based on "Overall Equipment Effectiveness" (OEE).

Integration

Times and quantities entered in BDE or MDE are accessed to display order progress.

Features

- Efficiency report
    - Number of units report and time-based efficiency report for all machines over an arbitrary choice of periods and shifts
    - Distribution graph showing rate of capacity utilization, scrap rate, rate of capacity utilization, technical efficiency and assignment utilization rate for several machines
- Performance profile
    - Evaluations of occupancy time, machine work time, assignment utilization rate and technical efficiency
- OEE report
    - Tabular evaluation of "Overall Equipment Effectiveness“ (OEE) to be able to consider the key performance indicators of productivity, quality and effectiveness for all machines. Graphical display of the OEE and its components.
    - Distribution graph showing OEE and its components over multiple machines
    - Tabular or graphical displays using bar or pie charts
    - Views of the top ten key performance indicators

2 Efficiency Report

Summary

Menu	Production facility management → Key figures → Performance report
Transaction code	effrp
Function authorization	effrp

The analysis concerns workplace/ machine-related performance data for a certain period of time and a certain number of workplaces. The result depends on the selection and therefore on the selection criteria made available on the selection panel. The performances displayed in graphic form for quantities and duration provide the production controller with the desired information immediately at a glance.

Selection criteria

The application provides the following selection criteria:

Workplace

This selection criterion references the workplace in the machine or workplace master data. You can also run a search using wildcards (placeholders *).

Group

This selection criterion references the group in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected group. You can also run a search using wildcards.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Cost center

This selection criterion references the cost center defined in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected cost center. You can also run a search using wildcards.

Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Designation

This selection criterion references the short name of the machines and workplaces in the machine's master data. Only those machines are displayed that are identical to the string that was entered. There is also the option to use wild cards (placeholders *) in this field.

Company

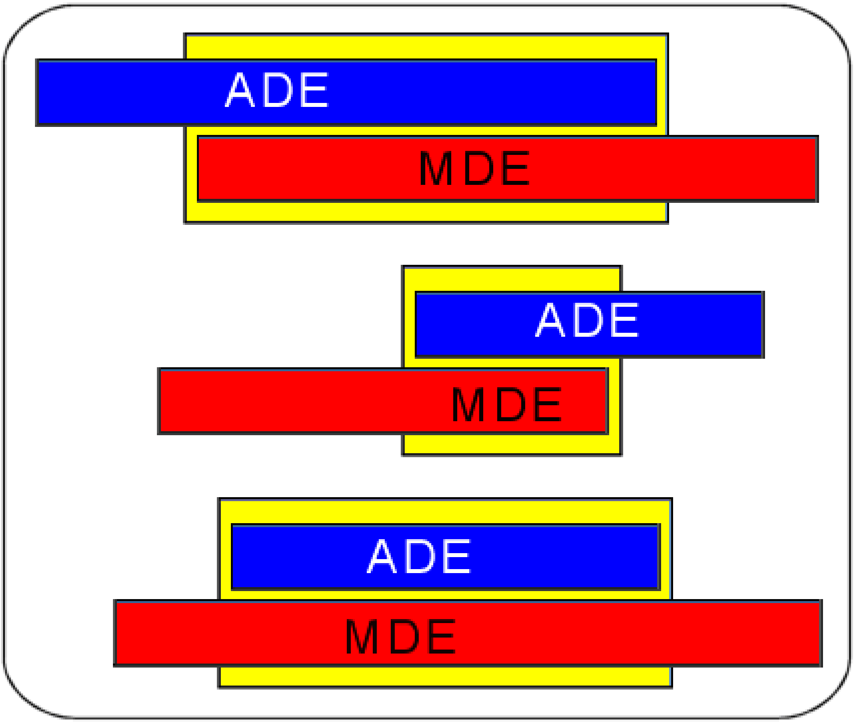
This selection criterion references the company defined in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected company. You can also run a search using wildcards.

Order/ article

For this kind of evaluation type, only the finished ADE postings are considered. If the order is currently still running on the machine, the time period between the last log on and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only ADE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the ADE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.

The order number, article/item or the batch (from the order header) may be used as selection criteria.

The following illustration shows an example of an overlapping of ADE and MDE postings. The ADE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity are assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Counter type

In the detail application "consumption figures", the counters to be displayed can be chosen by their type as defined within the configuration of counters. The detail application is only available if the relevant license has been purchased.

Efficiency report detail application

The efficiency report detail application includes workplace/ machine-related performance data for a certain period of time and a certain number of workplaces. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

***Workplace* category**

The following workplace/ machine-related master data are available:

- Workplace
- Short designation
- Designation
- Group
- Cost center
- Company

***Primary quantity, secondary quantity, tertiary quantity, basic quantity* category**

Workplace/ machine-related quantities recorded in the corresponding quantity types

- Yield
- Scrap
- Rework
- Open quantity

or the relevant quantity units (if relevant in the customer system).

***Duration* category**

- Production = RPA11

- Downtime = RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10

- Total = production + downtime

***Cycles* category**

- Number of posted cycles

***Key figures* category**

- Rate of capacity utilization
  The rate of capacity utilization represents the ratios derived from the effective runtime and the machine operation time.
  Rate of capacity utiliz. = 100/ (RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10 + RPA11) * RPA11

- Assignment utilization rate:
  The assignment utilization rate represents the relationship resulting from the effective runtime (main utilization time) and the adjusted machine operation time (i.e. without scheduled downtimes).
  Assignment utilization rate = 100 / (total of RPA 1 to RPA 11 minus RPA 6) * RPA 11

- Efficiency (efficiency is only available when combined with the "lines" license)
  Efficiency is the quotient derived from the effective runtime and the general runtime, meaning the sum total of the effective runtime and any interruptions as a result of machine-related disturbances or unscheduled shutdowns. Any other disturbances, e.g. organization-related disturbances, are not taken into account here.
  Efficiency = 100 / (RPA02 + RPA05 + RPA11) * RPA11

- Technical efficiency
  The variable showing the technical efficiency represents the sum total of the effective runtime and interruptions caused by technical (machine-related) disturbances. The times for all other disturbances (e.g. organization-related disturbances) are not accounted for in this calculation:
  Technical Efficiency = 100 / (RPA02 + RPA11) * RPA11

- Rate = 100 / ((yield (primary quantity) + scrap (primary quantity) + rework quantity (primary quantity) + open quantity (primary quantity)) * yield (primary quantity)

- Scrap rate = 100 / ((yield (primary quantity) + scrap (primary quantity)) * scrap (primary quantity)

Quantitative activities (machine-related) detail application

This detail application generates a graphic illustration of the quantities of the workplaces selected in the tabular detail application. Here, a differentiation is made by yield, scrap, rework and open quantity.

This detail application is docked right behind the quantity rate (group-related) detail application.

Quantity rate (group-related) detail application

This chart shows, based on the selected entries in the table, the relationship between the quantities - yield, scrap, rework and open quantities. The information is displayed by group/ added total by workplace group.

This detail application is docked right behind the quantitative activities (machine-related) detail application.

Duration detail application

This detail application generates a graphic illustration of the selected data from the efficiency report detail application. The durations are shown, broken down by RPA accounts.

You can hover the mouse over the graphic to display the value of the area where the mouse is. You have the option to switch between displaying the value in percent or in duration.

Consumption figures detail application

Shows the master data and counter values of the machine counters configured for the machines. By selecting the counter type, the data displayed can be limited to e.g. consumption figures or yield counters, or similar.

This detail application is only available if the system is configured accordingly and the relevant licenses are available.

3 Performance Profile

Summary

Menu	Production facility management → Key figures → Performance profile
Transaction code	effpf
Function authorization	effpf

The *Performance profile* application provides a tabular and graphic presentation of the production performance interrelationships. By compressing the performances entered by date and shift level, this application provides indispensable key production figures for all persons in a position of responsibility. This application supplies all quantities and durations at a glance that are necessary in order to be able to reliably assess the production status.

Selection criteria

The application provides the following selection criteria:

Workplace

This selection criterion references the workplace in the machine or workplace master data. You can also use wildcards (placeholders *).

Group

This selection criterion references the group in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected group. You can also run a search using wildcards.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Company

This selection criterion references the company defined in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected company. You can also run a search using wildcards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Cost center

This selection criterion references the cost center defined in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected cost center. You can also run a search using wildcards.

Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Designation

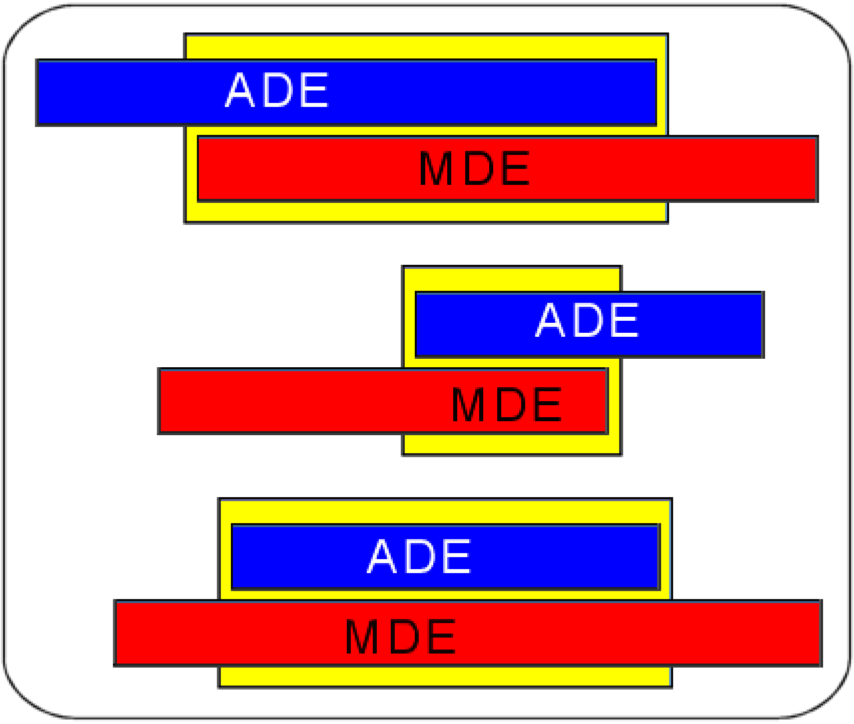
This selection criterion references the short name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders *) in this field.

Order/ article

For this kind of evaluation type, only the finished ADE postings are considered. If the order is currently still running on the machine, the time period between the last log on and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only ADE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the ADE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.

The order number, article/item or the batch (from the order header) may be used as selection criteria.

The following illustration shows an example of an overlapping of ADE and MDE postings. The ADE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity are assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Performance profile detail application

The performance profile of machine-related performance data is presented for a specific period of time and a certain number of workplaces. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

***Date* category**

Depending on the "Group results" option chosen in the selection range, the columns are filled as shown below (each of the columns not mentioned in the selection remain empty):

- Selection date and shift: Display year, calendar week, month, shift date, shift.
- Selection date: Display year, calendar week, month, shift date
- Selection week: Display year, calendar week
- Selection month: Display year, month

- Selection year: Display year

***Primary quantity, secondary quantity, tertiary quantity, basic quantity* category**

Workplace/ machine-related quantities recorded in the corresponding quantity types

- Yield
- Scrap
- Rework
- Open quantity

or the relevant quantity units (if relevant in the customer system).

***Duration* category**

- Production = RPA11

- Downtime = RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10

- Total = production + downtime

***Key figures* category**

- Rate of capacity utilization
  The rate of capacity utilization represents the ratios derived from the effective runtime and the machine operation time.
  Rate of capacity utiliz. = 100/ (RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10 + RPA11) * RPA11

- Assignment utilization rate:
  The assignment utilization rate represents the relationship resulting from the effective runtime (main utilization time) and the adjusted machine operation time (i.e. without scheduled downtimes).
  Assignment utilization rate = 100 / (total of RPA 1 to RPA 11 minus RPA 6) * RPA 11

- Efficiency (efficiency is only available when combined with the "lines" license)
  Efficiency is the quotient derived from the effective runtime and the general runtime, meaning the sum total of the effective runtime and any interruptions as a result of machine-related disturbances or unscheduled shutdowns. Any other disturbances, e.g. organization-related disturbances, are not taken into account here.
  Efficiency = 100 / (RPA02 + RPA05 + RPA11) * RPA11

- Technical efficiency
  The variable showing the technical efficiency represents the sum total of the effective runtime and interruptions caused by technical (machine-related) disturbances. The times for all other disturbances (e.g. organization-related disturbances) are not accounted for in this calculation:
  Technical Efficiency = 100 / (RPA02 + RPA11) * RPA11

- Rate = 100 / ((yield (primary quantity) + scrap
  (primary quantity) + rework quantity (primary quantity) + open quantity (primary quantity)) * yield (primary quantity)

- Scrap rate = 100 / ((yield (primary quantity) + scrap (primary quantity)) * scrap (primary quantity)

Chart quantities detail application

This chart shows, based on the selected entries in the table, the relationship between the quantities.

Depending on the option "group result" chosen in the selection range, the values are displayed per

- Shift/ shift date (selection date and shift)
- Shift date (selection date)
- Calendar week (selection week)
- Month (selection month)
- Year (selection year)

.

Using the "Displayed series" combo box, you can select which quantities you would like to have displayed (e.g. only yield and scrap).

Chart durations detail application

The durations detail application shows, based on the selected entries in the table, the relationship between the production duration to downtime. Depending on the option "group result" chosen in the selection range, the values are displayed per

- Shift/ shift date (selection date and shift)
- Shift date (selection date)

- Calendar week (selection week)
- Month (selection month)
- Year (selection year)

.

This detail application is docked right behind the key figures detail application.

Key figures chart detail application

The key figures detail application shows, based on the selected entries in the table, the efficiency or the technical efficiency, assignment utilization rate, rate, scrap rate or the rate of capacity utilization in graphic form.

To achieve this, the relevant key figure must be selected from the combo box. If no key figure is selected, the rate of capacity utilization is shown.

Depending on the option "group result" chosen in the selection range, the key figures are displayed per

- Shift/ shift date (selection date and shift)
- Shift date (selection date)
- Calendar week (selection week)
- Month (selection month)
- Year (selection year).

The key figure in the label (if activated) is displayed to the second decimal place.

This detail application is docked right behind the durations detail application.

4 OEE Report

Overview

Menu	Production facility/resource management --> Key performance indicators --> OEE report
Transaction code	oeerp
Function authorization	oeerp

The analysis concerns machine-related OEE performance data for a certain period of time and number of workplaces. The result depends on the selection made and therefore on the selection criteria available in the selection panel. This report lists expedient data in tabular form and provides sound information in graphic layouts. Consequently, it is a crucial tool for all users.

The application uses machine-related postings to calculate the KPIs.

Selection criteria

The application provides the following selection criteria:

Workplace

The selection criterion *workplace* refers to the workplace in the machine or workplace master data. You can also use wild cards (placeholders *).

Group

The selection criterion *group* refers to the group in the machine or workplace master data. The application shows all machines and/or workplaces that are assigned to the selected group. You can also use wild cards (placeholders *).

Date from ... to ...

The selection criterion *date from ... to ...* restricts the period you want to evaluate.

Shift / time

Use the selection criterion *shift* and/or *time* to further restrict the specified period (*date from .. to ...).* To do so, select a shift or specify a time (from ... until...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

If you select the "time" option, the application only includes those MDE log records that completely coincide with the selected period.

Report group

The selection criterion *report group* refers to the report groups. The application shows all machines or workplaces that are assigned to the selected report group.

Company

The selection criterion *company* refers to the company defined in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected company. You can also use wild cards (placeholders *).

Responsibility area

The selection criterion *responsibility area* refers to the responsibility area in the machine master data. Note that you can only view those machines you are authorized for (responsibility area).

Short name

The selection criterion *short name* refers to the short name of machines in the master data. The application shows all machines or workplaces matching the entered string. You can also use wild cards (placeholders *).

Cost center

The selection criterion *cost center* refers to the cost center stored in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected cost center. You can also use wild cards (placeholders *).

Name

The field *name* refers to the name of machines and workplaces defined in the machine master data. This field only shows the machines matching the entered string. You can also use wild cards (placeholders *).

Selection options

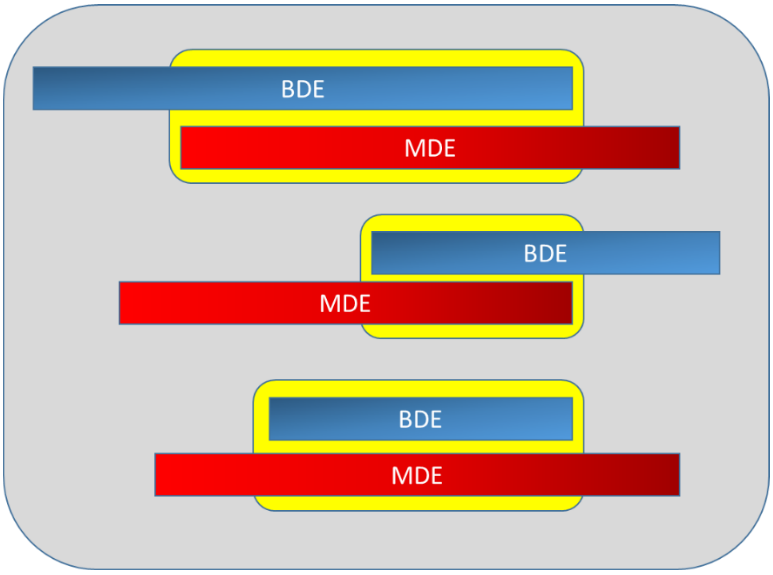
No selection:	If you choose this field, you cannot select the other selection criteria (order and resource).
Order:	Use order parameters to restrict the data.
Resource:	Use resource parameters to restrict the data.

If you check one of the options "order" or "resource", the application calculates the KPIs based on the machine-related postings that occurred while the order and/or resource was logged on. The application does not use the data posted onto the order and/or resource to calculate the KPIs. The application uses machine-related postings to calculate the KPIs.

Order/finished article/batch number

If you select the evaluation option *order,* the application identifies the BDE postings starting within the selection period and matching the entered selection criteria *order, finished article, batch number* (from the order header). If necessary, choose a selection period making sure that the required BDE postings actually coincide with this period.

The application compares these BDE postings and their periods with the machine-related postings. The below illustration shows an example of how BDE and MDE postings can overlap. The yellow areas represent the result of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If several orders are logged on to the machine simultaneously, this evaluation type (select one of the options *order, finished article* or *batch number)* assigns the full machine time (yellow area) and the quantities to every order. The fact that orders are run in parallel will not result in a proportionate calculation.

The application only includes completed BDE postings. If the operation is still logged on to the machine when you perform the evaluation, the time period between the last logon and now is not taken into account. Therefore, it is possible that differences appear between the machine evaluation and the order-related evaluation. For this evaluation type (select one of the options *order, finished article, batch number),* MPDV recommends selecting data via the "shift" option instead of the "time" option.

Resource/resource type

If you select the evaluation option *resource,* the application restricts the selected posting records integrated in the evaluation based on the logged in resources and/or the resources of a specific resource type. Then the application compares these posting records with the machine-related postings as described above.

OEE report detail application

The detail application *OEE report* refers to machine-related OEE performance data for a certain period of time and number of workplaces.

***Master data* category**

The following workplace/ machine-related master data are available:

- Workplace
- Short name
- Name
- Group
- Cost center
- Company

***Duration* category**

In addition to master data and the below-mentioned KPIs, the application shows the following data in tabular form:

- Planned operating time
- Machine runtime
- Actual utilization
- Yield utilization

How to calculate the individual KPIs is described further down in the document.

***Key performance indicators* category**

The OEE is calculated as follows:

OEE = Availability x Performance x Quality

How to calculate the individual KPIs is described further down in the document.

***Machine-related KPIs* detail application**

The diagram *machine-related KPIs* shows the following KPIs in a bar chart based on the machines selected in the table.

- OEE
- Availability
- Performance
- Quality

***Group-related KPIs* detail application**

The diagram *group-related KPIs* shows the following group-related KPIs in a bar chart based on the machines selected in the table.

- OEE
- Availability

- Performance
- Quality

Basic data detail application

The detail application *Basic data* displays basic data in graphic format

- Planned operating time
- Machine run time
- Actual utilization
- Yield utilization

for each workplace selected in the tabular detail application.

Values are durations, X-axis labeling is carried out automatically.

Formulas

OEE

OEE = Availability x Performance x Quality

Availability

Availability can be understood as a performance indicator of the machine. Like the OEE itself, it is a number less than one. Use the following formula to calculate the productivity of a machine for a specific period of time:

$$Availability = \frac{RPA11}{\sum_{1}^{11} RPA}$$

Formula: rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)
Use the formula **avail** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Performance

Use the following formula to calculate the performance of a machine for a specific period of time:

$$Performance = \frac{Target\ cycle^{*}}{Actual\ cycle}$$

Use the ratio of RPA 11 to the number of recorded cycles to identify the actual cycle.

$$Actual\ cycle = \left(\frac{\sum_{11} RPA}{\sum Cycles^{*}} \right)$$

The target cycle is an arithmetically averaged value, as different target cycles can be used within the selected period of time:

$$Target\ cycle^{*} = \frac{Target\ cycle\ arith.}{Duration\ arith.} = \frac{\sum (Target\ cycle * RPA11)}{\sum RPA11}$$

Cycles*: In case you could not collect cycles from the machine, calculate the cycles from the yield (primary quantity unit) and the partitioning:

$$Cycles^{*} = Yield_{Primary} / Partitioning$$

The application calculates the performance based on the MDE log records. The application uses all data records collected with the status "production" (data posted on RPA 11). The system calculates the performance for each log record. The application rates these individual performance values in order to show a compressed view in evaluations. This rating is based on the production time (RPA11).

$$Performance_{Product} = Performance * DURATION_{RPA11}$$

$$Performance = \frac{Performance_{Product}}{\sum RPA_{11}}$$

Example of how to calculate the performance with six MDE log records:

Machine	RPA11 [sec]	Performance	*Performance (Product)*
100	3600	0.7	2520
100	2700	0.8	2160
100	1800	0.9	1620
100	2700	0.5	1350
100	7200	0.9	6480
100	1800	0.9	1620
Total	19800	0.795	15750

Use the formula **pf_rat** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Quality

Quality represents the ratio of the produced yield to the total quantity (here: yield + scrap + rework + open quantity). This KPI provides information about the material to be processed and the quality of the process. Use the following formula to calculate the quality of a machine for a specific period of time:

$$Quality = \frac{Yield_{\text{Primary}}}{Yield_{\text{Primary}} + Scrap_{\text{Primary}} + \mathrm{Re}\,work_{\text{Primary}} + Open\_quantity_{\text{Primary}}}$$

Formula: yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary)
Use the formula **qual** to customize the calculation.
If the formula management does not include this formula, you have to create the formula in order to change the calculation.

NEE

In contrast to the OEE, NEE does not consider setup as a loss.

$$\frac{(RPA_{11} + RPA_7)}{\sum_{11}^{1} RPA} * Performance * Quality$$

Formula: ((rpa11 + rpa7) / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)) * performance * quality

Use the formula **nee** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Planned operating time

Total runtime of the machine during the selected period of time (Sum RPA 1 ... 11)

Formula: (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11)
Use the formula **op_ti** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Machine runtime

Main production time (RPA 11)

Formula: rpa11
Use the formula **mchf_rt** to customize the calculation.
If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Actual utilization

Yield utilization * (Target cycle / Actual cycle)
Yield utilization * Performance

Formula: rpa11 * (yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary)) * performance_rate
Use the formula **act_ut** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Yield utilization

Main production time * Quality

Formula: rpa11 * (yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary))
Use the formula **yie_ut** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.